

An Exact Reduction of the Karlsson Dita Circle to the Canonical Dittla Family

Consequences for Mutually Unbiased Bases and Certified Fourth-MUB Obstructions

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Abstract

We give an explicit matrix-level equivalence between the Dita specialization of Karlsson's three-parameter family of order-six complex Hadamard matrices and the canonical one-parameter Dittla family. On the Karlsson slice

$$\theta_D = \arccos(1/\sqrt{3}), \quad \phi_D = \pi/4,$$

the Möbius constraints reduce to

$$z_2^2 = z_3^2 = -1, \quad z_4^2 = z_1^{-2}.$$

Choosing the continuous branch

$$z_1 = e^{i\lambda}, \quad z_2 = i, \quad z_3 = -i, \quad z_4 = e^{-i\lambda}$$

produces a Laurent-polynomial representative $H_D(\lambda)$. We prove by direct matrix identity that

$$H_D(\lambda) = R P_r D\left(\frac{\lambda}{2\pi}\right) P_c,$$

where $D(x)$ is the canonical Dittla matrix, P_r, P_c are fixed permutation matrices, and R is a fixed diagonal unitary. Thus the parameter identification $x = \lambda/(2\pi)$ is explicit rather than merely an equivalence-class observation.

This reduction transfers mutually unbiased-basis (MUB) questions between the Karlsson and Dittla parameterizations. In particular, the equivalence relations $D(x) \simeq D(x + 1/2) \simeq D(-x + 1/4)$ induce $\lambda \simeq \lambda + \pi$ and $\lambda \simeq -\lambda + \pi/2$. We combine this exact reduction with the author's previously reported certified fourth-MUB obstructions at seven Karlsson Dita-slice parameters. The resulting statement remains pointwise: it does not by itself establish non-extendability on the entire continuous Dita circle. We identify the exact missing step needed for such a continuum theorem and formulate a parameterized witness-variety program for it.

1 Introduction

Mutually unbiased bases in dimension six remain an important computational and algebraic problem. If one member of an MUB pair is fixed to the standard basis, the second member is represented by a complex Hadamard matrix (CHM). Consequently, questions about triples and quadruples of MUBs can be studied through the geometry of order-six CHMs and their associated unbiased-vector varieties [2, 3, 1].

Karlssohn constructed a three-parameter family $K_6^{(3)}$ containing the known one- and two-parameter H_2 -reducible families as subfamilies [1]. A particularly important one-parameter specialization is obtained at

$$\theta_D = \arccos(1/\sqrt{3}), \quad \phi_D = \frac{\pi}{4}.$$

Karlssohn explicitly identifies this specialization as equivalent to the generic member of the Dittla family [1]. The present work makes that statement fully explicit at the matrix level and connects it directly to a certified computational MUB pipeline.

The author's preceding theorems manuscript established certified emptiness of the single-vector fourth-MUB witness W_1 for every recovered third-MUB 6-clique at seven points

$$\Lambda_{\text{cert}} = \left\{ 0, 0.4, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3} \right\},$$

with 40/40 recovered-pool cliques rejected by interval-certified homotopy continuation [5]. A companion computational report identified the full Dittla λ -circle as a reproducing third-MUB locus in the audited Karlssohn search, while explicitly distinguishing this computational identification from a classification theorem [6].

The purpose of this manuscript is therefore narrower and more algebraic:

- (i) derive the exact Karlssohn-to-Dittla parameter map;
- (ii) give the fixed permutations and diagonal rephasing explicitly;
- (iii) transport the known pointwise fourth-MUB certificates through that map;
- (iv) reduce the remaining continuum problem to a canonical Dittla parameter interval; and
- (v) formulate an exact parameterized witness-variety computation that could upgrade pointwise certificates to a continuum result.

2 Karlssohn's Dittla specialization

Karlssohn's H_2 -reducible family can be written in block form

$$H = \begin{pmatrix} F_2 & Z_1 & Z_2 \\ Z_3 & \frac{1}{2}Z_3AZ_1 & \frac{1}{2}Z_3BZ_2 \\ Z_4 & \frac{1}{2}Z_4BZ_1 & \frac{1}{2}Z_4AZ_2 \end{pmatrix}, \quad (2.1)$$

where

$$F_2 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

and

$$A = \begin{pmatrix} A_{11} & A_{12} \\ \overline{A}_{12} & -\overline{A}_{11} \end{pmatrix}, \quad B = -F_2 - A.$$

The entries relevant here are

$$A_{11} = -\frac{1}{2} + \frac{i\sqrt{3}}{2} (\cos \theta + e^{-i\phi} \sin \theta), \quad (2.2)$$

$$A_{12} = -\frac{1}{2} + \frac{i\sqrt{3}}{2} (-\cos \theta + e^{i\phi} \sin \theta). \quad (2.3)$$

At

$$\theta = \theta_D = \arccos(1/\sqrt{3}), \quad \phi = \phi_D = \frac{\pi}{4},$$

direct substitution gives

$$A = \begin{pmatrix} i & -1 \\ -1 & i \end{pmatrix}, \quad B = \begin{pmatrix} -1-i & 0 \\ 0 & 1-i \end{pmatrix}. \quad (2.4)$$

The Möbius constraints simplify to

$$z_2^2 = -1, \quad z_3^2 = -1, \quad z_4^2 = z_1^{-2}. \quad (2.5)$$

Choosing

$$z_1 = z = e^{i\lambda}, \quad z_2 = i, \quad z_3 = -i, \quad z_4 = \bar{z} \quad (2.6)$$

gives the dephased Laurent-polynomial representative

$$H_D(\lambda) = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & z & -z & i & -i \\ 1 & -i & i & i & -i & -1 \\ 1 & i & -z & z & -1 & -i \\ 1 & z^{-1} & -i & -1 & -z^{-1} & i \\ 1 & -z^{-1} & -1 & -i & z^{-1} & i \end{pmatrix}. \quad (2.7)$$

3 The canonical Dittia family

We use the standard dephased representative

$$D(x) = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & i & -i & -i & i \\ 1 & i & -1 & iz & -iz & -i \\ 1 & -i & i\bar{z} & -1 & i & -i\bar{z} \\ 1 & -i & -i\bar{z} & i & -1 & i\bar{z} \\ 1 & i & -i & -iz & iz & -1 \end{pmatrix}, \quad z = e^{2\pi ix}. \quad (3.1)$$

The canonical parameter satisfies

$$D(x) \simeq D(x + 1/2) \simeq D(-x + 1/4), \quad (3.2)$$

so a fundamental interval is

$$-\frac{1}{8} \leq x \leq \frac{1}{8}$$

[2].

4 Exact matrix-level equivalence

The central calculation of this work is the following.

Define row and column permutations by their index lists

$$r = (1, 6, 2, 3, 5, 4), \quad c = (2, 3, 5, 4, 6, 1),$$

where the entries specify the source row and column respectively. Let

$$R = \text{diag}(1, -i, -1, -i, i, i).$$

Thus, for a matrix M ,

$$(P_r M P_c)_{jk} = M_{r_j, c_k}.$$

Theorem 4.1 (Exact Karlsson–Dittl reduction). *For every real λ , with*

$$x = \frac{\lambda}{2\pi},$$

the Karlsson Dita specialization in equation (2.7) satisfies

$$\boxed{H_D(\lambda) = R P_r D\left(\frac{\lambda}{2\pi}\right) P_c.} \quad (4.1)$$

Proof. Set

$$z = e^{i\lambda} = e^{2\pi i x}.$$

Substitute this same z into equation (3.1). Applying the row permutation $r = (1, 6, 2, 3, 5, 4)$ and the column permutation $c = (2, 3, 5, 4, 6, 1)$, followed by multiplication of rows by

$$(1, -i, -1, -i, i, i),$$

gives exactly the six rows in equation (2.7). Every occurrence of z and z^{-1} is preserved under this operation. Hence the equality holds entrywise for all $z \in \mathbb{C}$ with $|z| = 1$, and therefore for all real λ . $\square$

Remark 4.2. The equivalence in equation (4.1) is stronger than numerical agreement of invariants. The permutations and diagonal unitary are independent of λ , so the entire one-parameter family is transported by one fixed equivalence transformation.

5 Induced parameter symmetries

Combining equation (4.1) with equation (3.2) immediately gives the following.

Corollary 5.1. *On the Karlsson Dita slice,*

$$H_D(\lambda) \simeq H_D(\lambda + \pi) \quad (5.1)$$

and

$$H_D(\lambda) \simeq H_D\left(-\lambda + \frac{\pi}{2}\right). \quad (5.2)$$

Consequently a fundamental parameter interval may be chosen as

$$-\frac{\pi}{4} \leq \lambda \leq \frac{\pi}{4}.$$

Proof. Under $x = \lambda/(2\pi)$, the transformations

$$x \mapsto x + \frac{1}{2}, \quad x \mapsto -x + \frac{1}{4}$$

become

$$\lambda \mapsto \lambda + \pi, \quad \lambda \mapsto -\lambda + \frac{\pi}{2}.$$

$\square$

For the seven previously certified parameters, the induced canonical Diltla coordinates are

λ	$x = \lambda/(2\pi)$	representative in $[-1/8, 1/8]$
0	0	0
0.4	$0.4/(2\pi)$	$0.4/(2\pi)$
$\pi/3$	$1/6$	$1/12$
$2\pi/3$	$1/3$	$-1/12$
π	$1/2$	0
$4\pi/3$	$2/3$	$1/12$
$5\pi/3$	$5/6$	$-1/12$

Thus the seven certified points represent four CHM equivalence classes.

6 Consequences for MUB extendability

Let

$$\mathcal{M}(H) = \{v \in \mathbb{C}^6 : |v_j|^2 = 1/6, |(H^\dagger v)_j|^2 = 1/6\}$$

denote the set of vectors unbiased to both the standard basis and H . For a third basis

$$B_3 = (v_1, \dots, v_6)$$

constructed from $\mathcal{M}(H)$, define the fourth-vector witness set

$$W_1(H, B_3) = \{w : w \text{ is MU to } I, H, B_3\}.$$

The author's preceding certified computation established

$$W_1(H_D(\lambda), B_3) = \emptyset$$

for every recovered-pool 6-clique at each

$$\lambda \in \Lambda_{\text{cert}},$$

with 40/40 cliques certified empty by interval-arithmetic certification of the homotopy-tracked witness system [5].

Because unitary equivalence preserves inner products, equation (4.1) transports each of those statements to the canonical Diltla parameterization.

Theorem 6.1 (Transport of the seven pointwise certificates). *For each*

$$\lambda \in \left\{0, 0.4, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}\right\},$$

the fourth-MUB obstruction established for the corresponding Karlsson matrix is equivalent to the corresponding obstruction for

$$D\left(\frac{\lambda}{2\pi}\right).$$

Remark 6.2. This theorem is deliberately pointwise. It does not assert that $W_1(D(x), B_3) = \emptyset$ for every $x \in [-1/8, 1/8]$. Nor does it classify all possible third-MUB bases at every parameter. Those are separate completeness questions.

7 Relation to the audited third-MUB locus

The companion computational report found the Dita λ -circle as a reproducing third-MUB locus in the audited Karlsson search. The report records 126/126 high-precision continuation probes on the Dita axis producing 6-cliques, while a coarse two-dimensional grid missed the locus because the relevant condition lies on the special $\phi = \pi/4$ slice [6]. At

$$H_D(0.4),$$

the same report records a certified per- H MU-pool computation with 240 distinct isolated roots against a mixed-volume bound of 252.

These computations support the geometric picture

$$\mathcal{V}_3 \longrightarrow \{\lambda\text{-circle}\},$$

where the fixed- H MU-vector system is zero-dimensional while the locus of third-MUB-compatible Hadamards is one-dimensional in parameter space.

The distinction is essential:

$$\underbrace{\dim \mathcal{M}(H_D(\lambda)) = 0}_{\text{fixed parameter}} \quad \text{versus} \quad \underbrace{\dim \mathcal{L}_3 = 1}_{\text{varying } \lambda}.$$

8 The remaining continuum problem

The exact reduction shows that the remaining question can be posed entirely inside the canonical Dittl family:

$$\boxed{\text{For every } x \in [-1/8, 1/8], \text{ does every third-MUB basis for } D(x) \text{ admit no fourth MU vector?}} \quad (8.1)$$

A direct attack should introduce the third-basis vectors as unknowns rather than selecting one numerically. After fixing the first component by projective gauge, write

$$v_a = \frac{1}{\sqrt{6}}(1, x_{a1}, \dots, x_{a5}), \quad a = 1, \dots, 5,$$

with inverse variables y_{aj} satisfying

$$x_{aj}y_{aj} - 1 = 0.$$

The equations impose:

$$|x_{aj}|^2 = 1, \quad (8.2)$$

$$|(D(x)^\dagger v_a)_j|^2 = \frac{1}{6}, \quad (8.3)$$

$$\langle v_a, v_b \rangle = 0, \quad a \neq b. \quad (8.4)$$

A fourth vector

$$w = \frac{1}{\sqrt{6}}(1, q_1, \dots, q_5)$$

is then added with equations

$$|q_j|^2 = 1,$$

$$|(D(x)^\dagger w)_j|^2 = \frac{1}{6},$$

and

$$|\langle v_a, w \rangle|^2 = \frac{1}{6}$$

for all a .

This defines a parameterized algebraic incidence variety

$$\mathcal{I}_4 = \{(x, B_3, w) : w \text{ is MU to } I, D(x), B_3\}.$$

The desired continuum theorem would follow from proving

$$\mathcal{I}_4 \cap ([-1/8, 1/8] \times \mathcal{B}_3) = \emptyset.$$

9 A certification strategy

Three complementary routes are available.

9.1 Parameter homotopy

Choose a certified anchor x_0 , enumerate the complete third-MUB pool, and track every solution branch as x varies. The existing pipeline already uses mixed-volume path tracking and interval certification at fixed anchors. The new step is to make x an explicit continuation parameter.

The critical events to monitor are:

- (a) Jacobian rank loss;
- (b) solution collisions;
- (c) changes in the number of MU vectors;
- (d) changes in the number of third-basis cliques;
- (e) singular witness solutions.

9.2 Symbolic parameter elimination

After clearing denominators and introducing $z = e^{2\pi i x}$, one can work over the Laurent polynomial ring

$$\mathbb{Q}(i)[z, z^{-1}, \mathbf{x}, \mathbf{y}, \mathbf{q}, \mathbf{r}].$$

Eliminating the vector variables gives an ideal in the parameter variable z . If the resulting elimination ideal contains a polynomial $P(z)$ whose zero set is disjoint from the unit circle, then no fourth-MUB witness exists on the full parameter circle.

The strongest possible outcome would be

$$1 \in I_{\text{elim}},$$

which would give an algebraic continuum obstruction.

9.3 Certified interval covering

If symbolic elimination remains computationally infeasible, partition

$$[-1/8, 1/8]$$

into certified intervals. For each interval, use parameter-dependent interval Newton/Krawczyk operators or a validated homotopy method to prove that no witness branch enters the interval.

This would upgrade the current finite set of point certificates to a continuum certificate without requiring a closed-form third basis.

10 New parameterized computations: continuation and the MU-vector fold

The exact reduction above permits a direct parameterized study in the canonical Dittla coordinate. Starting from a genuine third-MUB solution at $\lambda_0 = 0.4$, four distinct third-MUB bases were continued around the full 2π circle. The stored continuation data contain 516 branch states. The third-MUB residuals remain at the 10^{-15} scale, while the smallest sampled local Jacobian singular value remains positive; the lowest recorded values are approximately 9.4×10^{-2} . This is evidence that the selected branches are numerically smooth over the sampled circle. It is not a completeness theorem for the full third-MUB incidence variety.

A separate fourth-vector search used 17 parameter values on each of the four branches and 20 random starts at each state. No zero-residual fourth vector was found. The best recorded residuals lie approximately in

$$0.1443 \leq r_{\min} \leq 0.1795.$$

Because these are finite floating-point optimization searches, they are negative evidence only and do not replace the certified seven-point calculation.

A more structured computation revealed a transition in the recovered single-vector MU pool at

$$\lambda_* = 0.1114802243779665542913031975274\dots, \quad x_* = \frac{\lambda_*}{2\pi} = 0.0177426287667469900549425033849\dots \quad (10.1)$$

Immediately across a numerical bracket of width 4×10^{-6} around λ_* , the observed pool counts change from

$$N_v(\lambda_* - 2 \times 10^{-6}) = 120, \quad N_v(\lambda_* + 2 \times 10^{-6}) = 72.$$

The difference is $48 = 2 \times 24$.

To localize the singularity, the real phase-residual equations $F(\alpha, \lambda) = 0$ were augmented by

$$J_\alpha F(\alpha, \lambda)u = 0, \quad \|u\|_2 = 1.$$

High-precision computation recovered 24 distinct simultaneous singular MU-vector roots. A representative root has Jacobian singular values approximately

$$11.73, 9.70, 8.68, 4.65, 0,$$

and the numerical transversality quantities

$$w^T F_\lambda \approx -3.01808639127, \quad w^T F_{\alpha\alpha}[u, u] \approx -1.19270960464$$

are nonzero. These data are consistent with a simple fold normal form.

The fold is a singularity of the single-vector MU solution variety. It is not, by itself, a singularity of the selected six-vector third-MUB branches. In particular, a pool-level change in the number of individual MU vectors can occur while a chosen third-MUB continuation remains smooth.

No exact minimal polynomial for λ_* is claimed. High-precision PSLQ experiments produced candidate integer relations, but these have not been independently verified by modular elimination and exact substitution. They are therefore excluded from the theorem statements.

11 Reduced rational elimination and the remaining exact problem

For exact algebraic work it is natural to use the tangent-half-angle coordinate

$$z = \frac{1 + i\mu}{1 - i\mu}, \quad \mu = \tan(\lambda/2),$$

and, for each unknown phase,

$$x_j = \frac{1 + iq_j}{1 - iq_j}.$$

After taking real and imaginary parts and clearing denominators, the MU equations become polynomials over $\mathbb{Q}$. In the reduced single-vector system the first two MU equations are parameter-independent because the first two columns of the canonical Dittia family do not contain z . Their elimination should therefore precede any discriminant computation.

The preferred exact singular system is not the expanded determinant equation $\det J_q F = 0$, but the incidence system

$$F(q, \mu) = 0, \quad J_q F(q, \mu)u = 0, \quad u_5 = 1,$$

on a generic null-vector chart. Modular Groebner computations over several primes should be used to determine the elimination degree and factor pattern before attempting rational reconstruction. A numerical PSLQ relation from one root is not sufficient evidence for an exact polynomial.

The full fourth-MUB incidence variety is obtained by introducing five third-basis vectors

$$v_a = \frac{1}{\sqrt{6}}(1, x_{a1}, \dots, x_{a5}),$$

with inverse variables y_{aj} , together with a fourth vector

$$w = \frac{1}{\sqrt{6}}(1, q_1, \dots, q_5)$$

and inverse variables r_j . With $zt - 1 = 0$, the resulting ideal is

$$I_4 = \langle F_{\text{third}}, F_{\text{fourth}}, zt - 1, xy - 1, qr - 1 \rangle.$$

Eliminating all vector variables to $\mathbb{Q}(i)[z, t]$ and then imposing $zt = 1$ and $|z| = 1$ would provide the cleanest route to a continuum theorem. If global elimination is too expensive, validated parameter homotopy or an interval covering of the fundamental interval $\lambda \in [-\pi/4, \pi/4]$ is the appropriate fallback.

Table 1: Logical status of the combined exact, certified, and numerical results.

Statement	Status
Karlsson Dita specialization and Laurent representative	Exact algebra
Fixed permutation/rephasing relation to canonical Dittla family	Exact algebra
Parameter map $x = \lambda/(2\pi)$	Exact algebra
Seven pointwise fourth-MUB obstructions	Certified computation
Exact unit-ideal reference obstruction	Exact algebra / validation
Four selected third-MUB branches around 2π	High-precision numerical continuation
68 fourth-vector optimization states	Numerical negative evidence
Pool transition at λ_*	High-precision numerical localization
24 singular MU-vector roots	High-precision numerical computation
Simple-fold interpretation	Numerical local evidence
Exact polynomial for λ_*	Not established
Fourth-MUB absence on the entire Dita circle	Open
Fourth-MUB absence on all of $K_6^{(3)}$	Open
Zauner's conjecture $N(6) = 3$	Not addressed

12 Updated logical status

13 Conclusion and next exact attack

The Dita circle is now reduced to a single canonical parameter with an explicit exact matrix identity. The finite certified obstructions, full-circle third-MUB continuation, and the newly localized MU-vector fold all point to a structured algebraic geometry rather than an unstructured numerical search. The decisive next computation is an exact modular elimination of the reduced fold incidence system, followed by symmetry decomposition of the 24-root orbit. If a stable elimination polynomial is obtained, its unit-circle roots can be isolated exactly. The same strategy can then be lifted to the fourth-MUB incidence variety.

The present paper deliberately stops short of a continuum non-extendability theorem. That restraint is essential: the available computations establish exact identities, certified pointwise obstructions, and strong numerical structure, but they do not yet certify the absence of a fourth MUB for every parameter value on the circle.

14 Limitations and logical status

The results of this manuscript have deliberately different logical statuses.

In particular, the present reduction should not be interpreted as a proof of $N(6) = 3$. It concerns a single one-parameter subfamily of Karlsson's three-parameter family.

Table 2: Logical status of the main statements.

Statement	Status
Karlsson Dita specialization reduces to equation (2.7)	analytic
Exact fixed permutation/rephasing relation equation (4.1)	analytic
Induced Dittla parameter $x = \lambda/(2\pi)$	analytic
Seven pointwise fourth-MUB obstructions	certified, from prior work
Dita third-MUB locus on the audited search	high-precision computational finding
Fourth-MUB non-extendability for every x on the Dita circle	open
Non-extendability of all of $K_6^{(3)}$	open

15 Conclusion

The principal result is the explicit identity

$$H_D(\lambda) = R P_r D\left(\frac{\lambda}{2\pi}\right) P_c,$$

with

$$R = \text{diag}(1, -i, -1, -i, i, i),$$

$$r = (1, 6, 2, 3, 5, 4), \quad c = (2, 3, 5, 4, 6, 1).$$

This makes the Karlsson-to-Dittla correspondence completely explicit. The seven certified Karlsson Dita-slice obstructions therefore become seven canonical Dittla-family obstruction points, representing four CHM equivalence classes.

The important remaining problem is now sharply formulated: replace the finite set of certified parameter values by a certified statement over the fundamental Dittla interval. The natural next object is the parameterized fourth-MUB incidence variety $\mathcal{I}_4$, followed by continuation, singularity analysis, and, if possible, elimination in the single Laurent parameter z .

This gives a direct route from the author’s existing numerical and certified work to a genuinely stronger theorem without conflating computational identification with classification.

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